

# **WEEK 3 - ASSIGNMENT**

## **OBJECTIVE:**

Use Subqueries, CTEs, and Window Functions to analyze sales data from the Superstore dataset.

## **INTRODUCTION:**

This project analyzes the Superstore dataset using advanced SQL concepts such as Subqueries, Common Table Expressions (CTEs), and Window Functions. The objective is to derive customer sales insights, identify top-performing customers, analyze ordering behavior, and demonstrate the practical application of SQL analytical techniques for business decision-making.

## **DATASET:**

The Superstore dataset contains transactional sales records, including customer information, product details, order details, sales, profit, quantity, discount, and geographical information. It is widely used for sales and customer analytics. In this project, the dataset is used to analyze customer purchasing behavior, sales performance, customer rankings, and order patterns using advanced SQL techniques.

## **TASKS**

### **Step 1: Setup Data**

1. Import the Superstore dataset into a table named **superstore\_raw**
2. Create these 3 tables from it:
  - a. customers
  - b. products
  - c. orders
3. Insert data into these tables using SELECT DISTINCT

#### **customers table :**

**Query:**

```
CREATE TABLE customers AS
```

```

SELECT DISTINCT
    `Customer ID`,
    `Customer Name`,
    Segment
FROM superstore_raw;
SELECT * FROM customers;

```

### OUTPUT:

| Result Grid | Filter Rows:       | Export:     | Wrap Cell Content: |
|-------------|--------------------|-------------|--------------------|
| Customer ID | Customer Name      | Segment     |                    |
| CG-12520    | Claire Gute        | Consumer    |                    |
| DV-13045    | Darrin Van Huff    | Corporate   |                    |
| SO-20335    | Sean O'Donnell     | Consumer    |                    |
| BH-11710    | Brosina Hoffman    | Consumer    |                    |
| AA-10480    | Andrew Allen       | Consumer    |                    |
| IM-15070    | Irene Maddox       | Consumer    |                    |
| HP-14815    | Harold Pawlan      | Home Office |                    |
| PK-19075    | Pete Kriz          | Consumer    |                    |
| AG-10270    | Alejandro Grove    | Consumer    |                    |
| ZD-21925    | Zuschuss Donatelli | Consumer    |                    |
| KB-16585    | Ken Black          | Corporate   |                    |

### products table:

#### Query:

```

CREATE TABLE products AS
SELECT DISTINCT
    `Product ID`,
    `Product Name`,
    Category,
    `Sub-Category`
FROM superstore_raw;
SELECT * FROM products;

```

### OUTPUT:

| Result Grid     | Filter Rows:                                      | Export:         | Wrap Cell Content: | Fetch rows: |
|-----------------|---------------------------------------------------|-----------------|--------------------|-------------|
| Product ID      | Product Name                                      | Category        | Sub-Category       |             |
| FUR-BO-10001798 | Bush Somerset Collection Bookcase                 | Furniture       | Bookcases          |             |
| FUR-CH-10000454 | Hon Deluxe Fabric Upholstered Stacking Chairs,... | Furniture       | Chairs             |             |
| OFF-LA-10000240 | Self-Adhesive Address Labels for Typewriters b... | Office Supplies | Labels             |             |
| FUR-TA-10000577 | Bretford CR4500 Series Slim Rectangular Table     | Furniture       | Tables             |             |
| OFF-ST-10000760 | Eldon Fold 'N Roll Cart System                    | Office Supplies | Storage            |             |
| FUR-FU-10001487 | Eldon Expressions Wood and Plastic Desk Acces...  | Furniture       | Furnishings        |             |
| OFF-AR-10002833 | Newell 322                                        | Office Supplies | Art                |             |
| TEC-PH-10002275 | Mitel 5320 IP Phone VoIP phone                    | Technology      | Phones             |             |
| OFF-BI-10003910 | DXL Angle-View Binders with Locking Rings by S... | Office Supplies | Binders            |             |
| OFF-AP-10002892 | Belkin F5C206VTEL 6 Outlet Surge                  | Office Supplies | Appliances         |             |

### orders table:

#### **Query:**

```
CREATE TABLE orders AS
```

```
SELECT DISTINCT
```

```
  `Row ID`,
```

```
  `Order ID`,
```

```
  `Order Date`,
```

```
  `Ship Date`,
```

```
  `Customer ID`,
```

```
  `Product ID`,
```

```
  Sales,
```

```
  Quantity,
```

```
  Discount,
```

```
  Profit,
```

```
  Region,
```

```
  State,
```

```
  City
```

```
FROM superstore_raw;
```

```
SELECT * FROM orders LIMIT 10;
```

#### **OUTPUT:**

| Result Grid |        |                |            |            |             |                    |          |             |          |          |        |            |                 |
|-------------|--------|----------------|------------|------------|-------------|--------------------|----------|-------------|----------|----------|--------|------------|-----------------|
|             |        | Filter Rows:   |            | Export:    |             | Wrap Cell Content: |          | Fetch rows: |          |          |        |            |                 |
|             | Row ID | Order ID       | Order Date | Ship Date  | Customer ID | Product ID         | Sales    | Quantity    | Discount | Profit   | Region | State      | City            |
| ▶           | 1      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | CG-12520    | FUR-BO-10001798    | 261.96   | 2           | 0        | 41.9136  | South  | Kentucky   | Henderson       |
|             | 2      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | CG-12520    | FUR-CH-10000454    | 731.94   | 3           | 0        | 219.582  | South  | Kentucky   | Henderson       |
|             | 3      | CA-2016-138688 | 6/12/2016  | 6/16/2016  | DV-13045    | OFF-LA-10000240    | 14.62    | 2           | 0        | 6.8714   | West   | California | Los Angeles     |
|             | 4      | US-2015-108966 | 10/11/2015 | 10/18/2015 | SO-20335    | FUR-TA-10000577    | 957.5775 | 5           | 0.45     | -383.031 | South  | Florida    | Fort Lauderdale |
|             | 5      | US-2015-108966 | 10/11/2015 | 10/18/2015 | SO-20335    | OFF-ST-10000760    | 22.368   | 2           | 0.2      | 2.5164   | South  | Florida    | Fort Lauderdale |
|             | 6      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710    | FUR-FU-10001487    | 48.86    | 7           | 0        | 14.1694  | West   | California | Los Angeles     |
|             | 7      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710    | OFF-AR-10002833    | 7.28     | 4           | 0        | 1.9656   | West   | California | Los Angeles     |
|             | 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710    | TEC-PH-10002275    | 907.152  | 6           | 0.2      | 90.7152  | West   | California | Los Angeles     |
|             | 9      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710    | OFF-BI-10003910    | 18.504   | 3           | 0.2      | 5.7825   | West   | California | Los Angeles     |
|             | 10     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710    | OFF-AP-10002892    | 114.9    | 5           | 0        | 34.47    | West   | California | Los Angeles     |

## Step 2: Perform Required Queries

1. Find all orders where sales are greater than the average sales. (*Subquery*)

#### **Query:**

```
SELECT * FROM orders WHERE Sales > (SELECT AVG(Sales) FROM orders);
```

#### **Output:**

| Result Grid  |                |            |            |                    |                 |             |          |          |            |         |              |                 |  |
|--------------|----------------|------------|------------|--------------------|-----------------|-------------|----------|----------|------------|---------|--------------|-----------------|--|
| Filter Rows: |                | Export:    |            | Wrap Cell Content: |                 | Fetch rows: |          |          |            |         |              |                 |  |
| Row ID       | Order ID       | Order Date | Ship Date  | Customer ID        | Product ID      | Sales       | Quantity | Discount | Profit     | Region  | State        | City            |  |
| 1            | CA-2016-152156 | 11/8/2016  | 11/11/2016 | CG-12520           | FUR-BO-10001798 | 261.96      | 2        | 0        | 41.9136    | South   | Kentucky     | Henderson       |  |
| 2            | CA-2016-152156 | 11/8/2016  | 11/11/2016 | CG-12520           | FUR-CH-10000454 | 731.94      | 3        | 0        | 219.582    | South   | Kentucky     | Henderson       |  |
| 4            | US-2015-108966 | 10/11/2015 | 10/18/2015 | SO-20335           | FUR-TA-10000577 | 957.5775    | 5        | 0.45     | -383.031   | South   | Florida      | Fort Lauderdale |  |
| 8            | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710           | TEC-PH-10002275 | 907.152     | 6        | 0.2      | 90.7152    | West    | California   | Los Angeles     |  |
| 11           | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710           | FUR-TA-10001539 | 1706.184    | 9        | 0.2      | 85.3092    | West    | California   | Los Angeles     |  |
| 12           | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710           | TEC-PH-10002033 | 911.424     | 4        | 0.2      | 68.3568    | West    | California   | Los Angeles     |  |
| 14           | CA-2016-161389 | 12/5/2016  | 12/10/2016 | IM-15070           | TEC-PH-10003656 | 407.976     | 3        | 0.2      | 132.5922   | West    | Washington   | Seattle         |  |
| 17           | CA-2014-105893 | 11/11/2014 | 11/18/2014 | PK-19075           | OFF-ST-10004186 | 665.88      | 6        | 0        | 13.3176    | Central | Wisconsin    | Madison         |  |
| 25           | CA-2015-106320 | 9/25/2015  | 9/30/2015  | EB-13870           | FUR-TA-10000577 | 1044.63     | 3        | 0        | 240.2649   | West    | Utah         | Orem            |  |
| 28           | US-2015-150630 | 9/17/2015  | 9/21/2015  | TB-21520           | FUR-BO-10004834 | 3083.43     | 7        | 0.5      | -1665.0522 | East    | Pennsylvania | Philadelphia    |  |
| 36           | CA-2016-117590 | 12/8/2016  | 12/10/2016 | GH-14485           | TEC-PH-10004977 | 1097.544    | 7        | 0.2      | 123.4737   | Central | Texas        | Richardson      |  |
| 39           | CA-2015-117415 | 12/27/2015 | 12/31/2015 | SN-20710           | FUR-BO-10002545 | 532.3992    | 3        | 0.32     | -46.9764   | Central | Texas        | Houston         |  |
| 41           | CA-2015-117415 | 12/27/2015 | 12/31/2015 | SN-20710           | TEC-PH-10000486 | 371.168     | 4        | 0.2      | 41.7564    | Central | Texas        | Houston         |  |
| 55           | CA-2016-105816 | 12/11/2016 | 12/17/2016 | JM-15265           | TEC-PH-10002447 | 1029.95     | 5        | 0        | 298.6855   | East    | New York     | New York City   |  |
| 58           | CA-2016-111682 | 6/17/2016  | 6/18/2016  | TB-21055           | FUR-CH-10003968 | 319.41      | 5        | 0.1      | 7.098      | East    | New York     | Troy            |  |
| 68           | CA-2014-106376 | 12/5/2014  | 12/10/2014 | BS-11590           | OFF-AR-10002671 | 1113.024    | 8        | 0.2      | 111.3024   | West    | Arizona      | Gilbert         |  |

2. Find the highest sales order for each customer. (*Subquery*)

### Query:

```
SELECT o.* FROM orders o JOIN (SELECT `Customer ID`, MAX(Sales) AS Max_Sales
FROM orders GROUP BY `Customer ID`) m
ON o.`Customer ID` = m.`Customer ID`
AND o.Sales = m.Max_Sales;
```

### Output:

| Result Grid  |                |            |            |                    |                 |          |          |          |            |         |              |                 |  |
|--------------|----------------|------------|------------|--------------------|-----------------|----------|----------|----------|------------|---------|--------------|-----------------|--|
| Filter Rows: |                | Export:    |            | Wrap Cell Content: |                 |          |          |          |            |         |              |                 |  |
| Row ID       | Order ID       | Order Date | Ship Date  | Customer ID        | Product ID      | Sales    | Quantity | Discount | Profit     | Region  | State        | City            |  |
| 2            | CA-2016-152156 | 11/8/2016  | 11/11/2016 | CG-12520           | FUR-CH-10000454 | 731.94   | 3        | 0        | 219.582    | South   | Kentucky     | Henderson       |  |
| 4            | US-2015-108966 | 10/11/2015 | 10/18/2015 | SO-20335           | FUR-TA-10000577 | 957.5775 | 5        | 0.45     | -383.031   | South   | Florida      | Fort Lauderdale |  |
| 11           | CA-2014-115812 | 6/9/2014   | 6/14/2014  | BH-11710           | FUR-TA-10001539 | 1706.184 | 9        | 0.2      | 85.3092    | West    | California   | Los Angeles     |  |
| 25           | CA-2015-106320 | 9/25/2015  | 9/30/2015  | EB-13870           | FUR-TA-10000577 | 1044.63  | 3        | 0        | 240.2649   | West    | Utah         | Orem            |  |
| 28           | US-2015-150630 | 9/17/2015  | 9/21/2015  | TB-21520           | FUR-BO-10004834 | 3083.43  | 7        | 0.5      | -1665.0522 | East    | Pennsylvania | Philadelphia    |  |
| 36           | CA-2016-117590 | 12/8/2016  | 12/10/2016 | GH-14485           | TEC-PH-10004977 | 1097.544 | 7        | 0.2      | 123.4737   | Central | Texas        | Richardson      |  |
| 39           | CA-2015-117415 | 12/27/2015 | 12/31/2015 | SN-20710           | FUR-BO-10002545 | 532.3992 | 3        | 0.32     | -46.9764   | Central | Texas        | Houston         |  |
| 55           | CA-2016-105816 | 12/11/2016 | 12/17/2016 | JM-15265           | TEC-PH-10002447 | 1029.95  | 5        | 0        | 298.6855   | East    | New York     | New York City   |  |
| 58           | CA-2016-111682 | 6/17/2016  | 6/18/2016  | TB-21055           | FUR-CH-10003968 | 319.41   | 5        | 0.1      | 7.098      | East    | New York     | Troy            |  |
| 68           | CA-2014-106376 | 12/5/2014  | 12/10/2014 | BS-11590           | OFF-AR-10002671 | 1113.024 | 8        | 0.2      | 111.3024   | West    | Arizona      | Gilbert         |  |
| 110          | CA-2015-129476 | 10/15/2015 | 10/20/2015 | PA-19060           | TEC-AC-10000844 | 339.96   | 5        | 0.2      | 67.992     | Central | Illinois     | Orland Park     |  |
| 146          | CA-2015-110744 | 9/7/2015   | 9/12/2015  | HA-14920           | OFF-ST-10003656 | 671.93   | 7        | 0        | 20.1579    | West    | California   | Pasadena        |  |

3. Calculate total sales for each customer. (*CTE*)

### Query:

```
WITH customer_sales AS (SELECT `Customer ID`, SUM(Sales) AS Total_Sales FROM orders
GROUP BY `Customer ID`)
SELECT * FROM customer_sales;
```

### Output:

| Result Grid                                                                                                                                                                                      |             |                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|
| Filter Rows: <input type="text"/>                                                                                                                                                                |             |                              |
| Export:  Wrap Cell Content:  |             |                              |
|                                                                                                                                                                                                  | Customer ID | Total_Sales                  |
| ▶                                                                                                                                                                                                | CG-12520    | 1148.7800000000002           |
|                                                                                                                                                                                                  | DV-13045    | 1119.4830000000002           |
|                                                                                                                                                                                                  | SO-20335    | 2602.5755 6255.3510000000001 |
|                                                                                                                                                                                                  | BH-11710    | 6255.3510000000001           |
|                                                                                                                                                                                                  | AA-10480    | 1782.872                     |
|                                                                                                                                                                                                  | IM-15070    | 4930.4739999999999           |
|                                                                                                                                                                                                  | HP-14815    | 1990.314                     |
|                                                                                                                                                                                                  | PK-19075    | 8158.654                     |
|                                                                                                                                                                                                  | AG-10270    | 2565.2579999999994           |
|                                                                                                                                                                                                  | ZD-21925    | 1493.944                     |
|                                                                                                                                                                                                  | KB-16585    | 2742.7580000000003           |
|                                                                                                                                                                                                  | SF-20065    | 763.5459999999998            |
|                                                                                                                                                                                                  | EB-13870    | 2767.219                     |

4. Find customers whose total sales are above average. (CTE + Subquery)

**Query:**

WITH customer\_sales AS (SELECT `Customer ID`, SUM(Sales) AS Total\_Sales FROM orders  
GROUP BY `Customer ID`)

SELECT \* FROM customer\_sales WHERE Total\_Sales > (SELECT AVG(Total\_Sales) FROM  
customer\_sales);

**Output:**

| Result Grid                                                                                                                                                                                          |             |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|
| Filter Rows: <input type="text"/>                                                                                                                                                                    |             |                    |
| Export:  Wrap Cell Content:  |             |                    |
|                                                                                                                                                                                                      | Customer ID | Total_Sales        |
| ▶                                                                                                                                                                                                    | BH-11710    | 6255.3510000000001 |
|                                                                                                                                                                                                      | IM-15070    | 4930.4739999999999 |
|                                                                                                                                                                                                      | PK-19075    | 8158.654           |
|                                                                                                                                                                                                      | TB-21520    | 4730.628           |
|                                                                                                                                                                                                      | MA-17560    | 4280.6209999999999 |
|                                                                                                                                                                                                      | SN-20710    | 3127.9592000000007 |
|                                                                                                                                                                                                      | LC-16930    | 4481.162           |
|                                                                                                                                                                                                      | RA-19885    | 3832.3139999999994 |
|                                                                                                                                                                                                      | ES-14080    | 4648.516           |
|                                                                                                                                                                                                      | KM-16720    | 4909.4720000000001 |
|                                                                                                                                                                                                      | KD-16270    | 8282.358           |
|                                                                                                                                                                                                      | JE-15745    | 6400.4549999999999 |

5. Rank all customers based on total sales. (Window Function)

**Query:**

WITH customer\_sales AS (SELECT `Customer ID`, SUM(Sales) AS Total\_Sales FROM orders  
GROUP BY `Customer ID`)

SELECT `Customer ID`, Total\_Sales, RANK() OVER(ORDER BY Total\_Sales DESC) AS  
Sales\_Rank FROM customer\_sales;

**Output:**

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content: |             |                    |            |
|----------------------------------------------------------------------------------|-------------|--------------------|------------|
|                                                                                  | Customer ID | Total_Sales        | Sales_Rank |
| ▶                                                                                | SM-20320    | 25043.05           | 1          |
|                                                                                  | TC-20980    | 19017.847999999998 | 2          |
|                                                                                  | RB-19360    | 15117.339          | 3          |
|                                                                                  | TA-21385    | 14595.62           | 4          |
|                                                                                  | AB-10105    | 14355.610999999997 | 5          |
|                                                                                  | SC-20095    | 14142.333999999999 | 6          |
|                                                                                  | KL-16645    | 14071.917          | 7          |
|                                                                                  | HL-15040    | 12873.297999999999 | 8          |
|                                                                                  | SE-20110    | 12209.438000000002 | 9          |
|                                                                                  | CC-12370    | 12129.072          | 10         |
|                                                                                  | TS-21370    | 11885.871          | 11         |

6. Assign row numbers to each order within a customer. (*Window Function + PARTITION BY*)

**Query:**

SELECT `Customer ID`, `Order ID`, `Order Date`, Sales, ROW\_NUMBER() OVER (PARTITION BY `Customer ID` ORDER BY `Order Date`) AS Order\_Number FROM orders;

**Output:**

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content: |             |                |            |          |              |
|----------------------------------------------------------------------------------|-------------|----------------|------------|----------|--------------|
|                                                                                  | Customer ID | Order ID       | Order Date | Sales    | Order_Number |
| ▶                                                                                | AA-10315    | CA-2015-121391 | 10/4/2015  | 26.96    | 1            |
|                                                                                  | AA-10315    | CA-2016-103982 | 3/3/2016   | 3930.072 | 2            |
|                                                                                  | AA-10315    | CA-2016-103982 | 3/3/2016   | 2.304    | 3            |
|                                                                                  | AA-10315    | CA-2016-103982 | 3/3/2016   | 431.976  | 4            |
|                                                                                  | AA-10315    | CA-2016-103982 | 3/3/2016   | 41.72    | 5            |
|                                                                                  | AA-10315    | CA-2014-128055 | 3/31/2014  | 673.568  | 6            |
|                                                                                  | AA-10315    | CA-2014-128055 | 3/31/2014  | 52.98    | 7            |
|                                                                                  | AA-10315    | CA-2017-147039 | 6/29/2017  | 362.94   | 8            |
|                                                                                  | AA-10315    | CA-2017-147039 | 6/29/2017  | 11.54    | 9            |
|                                                                                  | AA-10315    | CA-2014-138100 | 9/15/2014  | 14.56    | 10           |
|                                                                                  | AA-10315    | CA-2014-138100 | 9/15/2014  | 14.94    | 11           |
|                                                                                  | AA-10375    | CA-2014-130729 | 10/24/2014 | 34.272   | 1            |
|                                                                                  | AA-10375    | CA-2015-114503 | 11/13/2015 | 84.96    | 2            |

7. Display top 3 customers based on total sales. (*Window Function*)

**Query:**

WITH customer\_sales AS (SELECT `Customer ID`, SUM(Sales) AS Total\_Sales FROM orders GROUP BY `Customer ID`)

SELECT \* FROM (SELECT `Customer ID`, Total\_Sales, RANK() OVER(ORDER BY Total\_Sales DESC) AS Sales\_Rank FROM customer\_sales ) ranked\_customers

WHERE Sales\_Rank <= 3;

### **Output:**

| Result Grid | Filter Rows:       | Export:    | Wrap Cell Content: |
|-------------|--------------------|------------|--------------------|
| Customer ID | Total_Sales        | Sales_Rank |                    |
| SM-20320    | 25043.05           | 1          |                    |
| TC-20980    | 19017.847999999998 | 2          |                    |
| RB-19360    | 15117.339          | 3          |                    |

### **Step 3: Final Combined Query**

One final query that shows:

- Customer Name
- Total Sales
- Rank

(Use JOIN + CTE + Window Function together)

### **Query:**

WITH customer\_sales AS (SELECT `Customer ID`, SUM(Sales) AS Total\_Sales FROM orders GROUP BY `Customer ID`)

SELECT c.`Customer ID`, c.`Customer Name`, cs.Total\_Sales, RANK() OVER(ORDER BY cs.Total\_Sales DESC) AS Sales\_Rank FROM customer\_sales cs JOIN customers c ON cs.`Customer ID` = c.`Customer ID` ORDER BY Sales\_Rank;

### **Output:**

| Result Grid | Filter Rows:       | Export:            | Wrap Cell Content: |
|-------------|--------------------|--------------------|--------------------|
| Customer ID | Customer Name      | Total_Sales        | Sales_Rank         |
| SM-20320    | Sean Miller        | 25043.05           | 1                  |
| TC-20980    | Tamara Chand       | 19017.847999999998 | 2                  |
| RB-19360    | Raymond Buch       | 15117.339          | 3                  |
| TA-21385    | Tom Ashbrook       | 14595.62           | 4                  |
| AB-10105    | Adrian Barton      | 14355.610999999997 | 5                  |
| SC-20095    | Sanjit Chand       | 14142.333999999999 | 6                  |
| KL-16645    | Ken Lonsdale       | 14071.917          | 7                  |
| HL-15040    | Hunter Lopez       | 12873.297999999999 | 8                  |
| SE-20110    | Sanjit Engle       | 12209.438000000002 | 9                  |
| CC-12370    | Christopher Conant | 12129.072          | 10                 |
| TS-21370    | Todd Sumrall       | 11885.871          | 11                 |
| GT-14710    | Greg Tran          | 11820.119999999997 | 12                 |
| BM-11140    | Becky Martin       | 11609.900000000001 | 13                 |
| SV-20365    | Seth Vernon        | 11420.645999999997 | 14                 |
| CJ-12010    | Caroline Jumper    | 11079.742          | 15                 |
| CL-12565    | Clay Ludtke        | 10880.545999999998 | 16                 |

# MINI PROJECT: Customer Sales Insights

## 1. Who are the top 5 customers?

### Query:

```
WITH customer_sales AS (SELECT c.`Customer Name`, SUM(o.Sales) AS Total_Sales
FROM orders o
JOIN customers c
ON o.`Customer ID` = c.`Customer ID`
GROUP BY c.`Customer Name`
)
SELECT * FROM customer_sales ORDER BY Total_Sales DESC LIMIT 5;
```

### Output:

| Result Grid | Filter Rows:  | Export:            | Wrap Cell Content: |
|-------------|---------------|--------------------|--------------------|
|             | Customer Name | Total_Sales        |                    |
| ▶           | Sean Miller   | 25043.05           |                    |
|             | Tamara Chand  | 19017.847999999998 |                    |
|             | Raymond Buch  | 15117.339          |                    |
|             | Tom Ashbrook  | 14595.62           |                    |
|             | Adrian Barton | 14355.610999999997 |                    |

## 2. Who are the bottom 5 customers?

### Query:

```
WITH customer_sales AS ( SELECT c.`Customer Name`, SUM(o.Sales) AS Total_Sales
FROM orders o
JOIN customers c
ON o.`Customer ID` = c.`Customer ID`
GROUP BY c.`Customer Name`
)
SELECT * FROM customer_sales ORDER BY Total_Sales ASC LIMIT 5;
```

### Output:

| Result Grid | Filter Rows:    | Export:     | Wrap Cell Content: |
|-------------|-----------------|-------------|--------------------|
|             | Customer Name   | Total_Sales |                    |
| ▶           | Thais Sissman   | 4.833       |                    |
|             | Lela Donovan    | 5.304       |                    |
|             | Mitch Gastineau | 12.32       |                    |
|             | Carl Jackson    | 16.52       |                    |
|             | Roy Skaria      | 22.328      |                    |

### 3. Which customers made only one order?

#### Query:

```
SELECT
    c.`Customer Name`,
    COUNT(DISTINCT o.`Order ID`) AS Order_Count
FROM orders o
JOIN customers c
ON o.`Customer ID` = c.`Customer ID`
GROUP BY c.`Customer Name`
HAVING COUNT(DISTINCT o.`Order ID`) = 1;
```

#### Output:

| Result Grid |                   |             | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-------------------|-------------|--------------|---------|--------------------|
|             | Customer Name     | Order_Count |              |         |                    |
| ▶           | Anemone Ratner    | 1           |              |         |                    |
|             | Anthony O'Donnell | 1           |              |         |                    |
|             | Carl Jackson      | 1           |              |         |                    |
|             | Jenna Caffey      | 1           |              |         |                    |
|             | Jocasta Rupert    | 1           |              |         |                    |
|             | Lela Donovan      | 1           |              |         |                    |
|             | Mitch Gastineau   | 1           |              |         |                    |
|             | Patricia Hirasaki | 1           |              |         |                    |
|             | Ricardo Emerson   | 1           |              |         |                    |
|             | Roland Murray     | 1           |              |         |                    |
|             | Susan MacKendrick | 1           |              |         |                    |
|             | Theresa Coyne     | 1           |              |         |                    |

### 4. Which customers have above-average sales?

#### Query:

```
WITH customer_sales AS (
    SELECT
        c.`Customer Name`,
        SUM(o.Sales) AS Total_Sales
    FROM orders o
    JOIN customers c
    ON o.`Customer ID` = c.`Customer ID`
    GROUP BY c.`Customer Name`
)
SELECT * FROM customer_sales WHERE Total_Sales > (SELECT AVG(Total_Sales)
FROM customer_sales);
```

#### Output:

| Result Grid |                    |                    | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------------------|--------------------|--------------|---------|--------------------|
|             | Customer Name      | Total_Sales        |              |         |                    |
| ▶           | Brosina Hoffman    | 6255.351000000001  |              |         |                    |
|             | Irene Maddox       | 4930.473999999999  |              |         |                    |
|             | Pete Kriz          | 8158.654           |              |         |                    |
|             | Tracy Blumstein    | 4730.628           |              |         |                    |
|             | Matt Abelman       | 4280.620999999999  |              |         |                    |
|             | Steve Nguyen       | 3127.9592000000007 |              |         |                    |
|             | Linda Cazamias     | 4481.162           |              |         |                    |
|             | Ruben Ausman       | 3832.3139999999994 |              |         |                    |
|             | Erin Smith         | 4648.516           |              |         |                    |
|             | Kunst Miller       | 4909.472000000001  |              |         |                    |
|             | Karen Daniels      | 8282.358           |              |         |                    |
|             | Joel Eaton         | 6400.454999999999  |              |         |                    |
|             | Stewart Carmic...  | 4492.656999999999  |              |         |                    |
|             | Christopher Schild | 3651.8585000000007 |              |         |                    |

## 5. What is the highest order value per customer?

### Query:

SELECT

c.`Customer ID`,

c.`Customer Name`,

MAX(o.Sales) AS Highest\_Order\_Value

FROM orders o

JOIN customers c

ON o.`Customer ID` = c.`Customer ID`

GROUP BY c.`Customer ID`, c.`Customer Name`

ORDER BY Highest\_Order\_Value DESC;

### Output:

| Result Grid |             |                    |                     | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-------------|--------------------|---------------------|--------------|---------|--------------------|
|             | Customer ID | Customer Name      | Highest_Order_Value |              |         |                    |
| ▶           | SM-20320    | Sean Miller        | 22638.48            |              |         |                    |
|             | TC-20980    | Tamara Chand       | 17499.95            |              |         |                    |
|             | RB-19360    | Raymond Buch       | 13999.96            |              |         |                    |
|             | TA-21385    | Tom Ashbrook       | 11199.968           |              |         |                    |
|             | HL-15040    | Hunter Lopez       | 10499.97            |              |         |                    |
|             | AB-10105    | Adrian Barton      | 9892.74             |              |         |                    |
|             | SC-20095    | Sanjit Chand       | 9449.95             |              |         |                    |
|             | BS-11365    | Bill Shonely       | 9099.93             |              |         |                    |
|             | SE-20110    | Sanjit Engle       | 8749.95             |              |         |                    |
|             | CC-12370    | Christopher Conant | 8399.976            |              |         |                    |
|             | KL-16645    | Ken Lonsdale       | 8187.65             |              |         |                    |
|             | BM-11140    | Becky Martin       | 8159.952            |              |         |                    |
|             | GT-14635    | Grant Thornton     | 7999.98             |              |         |                    |
|             | TB-21400    | Tom Boeckenhauer   | 6999.96             |              |         |                    |

## **BUSINESS INSIGHTS :**

1. Customer sales are unevenly distributed, with a small group of customers contributing a significant share of total revenue.
2. The Top 5 customers generate the highest sales and can be targeted through loyalty programs, exclusive offers, and personalized services.
3. Several customers have placed only one order, highlighting opportunities for customer retention and repeat-purchase campaigns.
4. Customers with above-average sales represent high-value customers and should be prioritized in marketing and relationship management strategies.
5. Customer ranking using window functions helps identify top-performing customers and supports data-driven decision-making.
6. Highest-order-value analysis identifies customers who make large purchases and contribute significantly to business revenue.
7. Order-level analysis using ROW\_NUMBER() helps track customer purchasing patterns and order history.
8. CTEs and Subqueries simplify complex analytical tasks and improve query readability and maintainability.

## **CONCLUSION:**

This project successfully applied Subqueries, CTEs, and Window Functions to analyze Superstore sales data. Through customer ranking, sales aggregation, and order analysis, meaningful business insights were generated. The project demonstrates how advanced SQL techniques can be used to support data-driven business decisions and customer analytics.